



Prioritization of Agile Requirement Elicitation Techniques Using Fuzzy Logic

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Abstract As Agile software development provides a more flexible and adaptable approach to developing software, its need has increased. In the context of requirement elicitation, agile plays a crucial role in identifying and understanding user requirements; however, there are many techniques available, and selecting the right one for your project can be difficult. In this research, we have proposed an approach using Fuzzy logic to prioritize such techniques. It incorporates factors such as ease of use of a technique, how frequently it is used by practitioners, and collaboration. This approach helps agile teams to make more informed decisions about which requirement elicitation technique they should use for their project based on how widely an approach is being used by practitioners. The result of our study indicated that Use Case, User Story, Prototype, Scenario, and UML Diagram are techniques widely implemented in relation to the overall importance weight based on the three factors.

Keywords Requirements engineering · RQ · Extreme programming · Fuzzy logic · Research methodology · User interface

1 Introduction

Requirements engineering (RE) is the most important phase of the software development life cycle (SDLC). The goal of requirements engineering is to turn the unclear, ad hoc requirements, and aspirations of potential customers into unambiguous, formal specifications. The specifications serve as a contract between the developer and the user. As a result, Requirement Engineering is essential in the development of effective software and the prevention of software errors early on [1]. It is widely known that problems with requirements engineering (RE) can affect the effectiveness of the software development process as well as the quality of the software. Addressing requirements problems later (e.g., during maintenance) is predicted to be 200 times more expensive than correcting them during the requirements phase. Inadequate requirements have caused many software projects to fail. If requirements are not well documented and managed, no software process can keep delivery schedules, budgets, or product quality within control [2].

Software engineers have been debating how to arrange software development in order to provide faster, better, and less expensive solutions for decades. There have been numerous suggestions for improvement, ranging from software process standardization and measurement to a variety of concrete tools, methodologies, and practices [3]. Agile software development methodologies are making an effort to satisfy the needs of a corporate community that is hungry for software development processes that are more lightweight, quick, and nimble. “Ready for motion, nimbleness, activity, skill in motion” is what agile means. This is particularly true in the unpredictable and quickly expanding internet software industry and the increasing mobile app market. Several discussions and studies have been sparked by the emerging agile methodologies [4]. Agile fulfills modern-

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day business requirements by incorporating proactive and reactive changes based on trust, proprietorship, continuous improvements, learnability, and willingly adopting openness and growth [5].

One of the knowledge-intensive procedures in the software development process is the requirement elicitation process, which is critical to the success of the software system. The goal of the requirement elicitation technique is to figure out what the user wants or needs. Requirement elicitation methodologies are typically chosen based on company policy or personal experience. There is also some guidance on selecting elicitation approaches for a new software project [6].

In RE, requirements elicitation is used to define the system's behavior and establish its boundaries. The identification of stakeholders (end users, customers, developers, and sponsors) and understanding of their needs are essential to success. The right approach choice is essential for eliciting requirements. Literature has already emphasized the significance of technique selection in the requirement elicitation process. There have been many models for requirement elicitation that have focused on technique selection. Many of the models that have been discussed in the literature have the drawback of requiring human involvement during technique selection. The decision to choose a technique may be biased by this human involvement. Some agile elicitation techniques are User Stories, Prototypes, Story Cards, UML diagrams, Kanban Board, UI patterns, and Storyboard Tasks.

This paper focuses on constructing a formula to prioritize requirement elicitation techniques based on the basis of three parameters using fuzzy logic. Our contributions in this paper include

1. We identified what requirement elicitation techniques are used in agile.
2. We proposed a model using fuzzy logic on the data collected from the survey to prioritize requirement elicitation techniques in agile.

The paper consists of the following sections: Sect. 1 begins with an overview of requirement elicitation (RE) and its function in Agile. Section 2 gives a quick rundown of relevant work. We prioritize agile requirement elicitation techniques using fuzzy logic in Sect. 3. In Sect. 4, the results of prioritization are discussed. Section 5 concludes with a brief summary.

2 Related Work

Requirement elicitation is the process of establishing stakeholders' demands and organizing this data in a comprehensible way so that developers may create a system that

will meet those needs [6]. Since the requirement elicitation is quite mature but still considered a critical phase in requirement engineering. Elicitation approaches might be conventional, collaborative, innovative, and contextual. One elicitation technique is insufficient depending on the situation and the nature of the project [7]. Numerous types of research have been done in-depth regarding requirement elicitation techniques [8,9]. As there is a vast variety of different techniques and approaches available, there is no prioritization.

In the software development industry, user demands, emerging technology, and increased complexity evolve rapidly and agility becomes a need. The agile approach is the one that is used the most in software development to deal with the inevitable changes in requirements. This study includes a discussion of the elicitation techniques that are employed during the agile development phase. However, each technique has its own limitations and makes assumptions regarding the project [10].

In agile software development, requirements are accepted late so it is challenging to prioritize them. In the [11] study, the importance of requirement prioritization is highlighted. The authors have constructed a prioritization repository and then provided a detailed analysis of the prioritization techniques based on their strengths and weaknesses. There are multiple requirement prioritization techniques that do not handle requirement growth and dependencies in requirements. The authors in [12] introduced a framework to make the prioritization technique more continuous and scalable to handle changes and dependencies in requirements.

Takeuchi et al. [13] acknowledged in the middle of the 1980s that a sequential stages approach to product development is unproductive because it lacks flexibility. New process models have since been created. There are iterative process models like the Rational Unified Process on the one side [13,14]. Scrum [15,16], Extreme Programming (XP) [17], Feature-Driven Development [18], and Kanban are examples of agile methodologies. Between 2004 and 2010, Anderson had an impact on the implementation of Kanban in Information Technology (IT) [19]. There are many published studies on lightweight process models.

Unlike agile methodologies, which begin with a closed phase, RE is done incrementally throughout the development process. The primary objective of Inayat et al. [20]'s literature review is to increase awareness of Agile RE issues and practices. They also wanted to know how Agile RE is used to overcome typical RE difficulties. In conclusion, they offered a few common practices as well as real obstacles that agile teams had to deal with. User stories, iterative requirements, requirement prioritization, change management, cross-functional teams, prototyping, testing before coding, requirements modeling, requirements management, review meetings, acceptance tests, code refactoring, shared conceptualization, pairing for requirements analysis, retro-

spectives, and continuous planning are a few of the practices used.

The authors in [21] employ the systematic literature review to provide an understanding of the requirement prioritization techniques and their empirical evaluation based on ease of use, effectiveness, and scalability. The findings indicate that fuzzy logic is frequently utilized in the prioritization such as in [22,23]. Table 1 gives the details of utilization of fuzzy methods in requirement prioritization.

Fuzzy logic has been utilized to evaluate agile methods. In a recent study, fuzzy logic is utilized to prioritize distributed agile patterns [33]. Fuzzy logic is utilized with prioritization of requirement elicitation techniques to control ambiguity and uncertainty. When trying to prioritize a list, there can be a lot of different factors to consider, and fuzzy logic deals with such ambiguities. This literature review covers a wide range of Agile Requirement Elicitation techniques. However, many researchers are working to identify agile requirements elicitation techniques, but none of them can say, which is the best requirement elicitation technique that fits all projects and is the most reliable based on ease of use, frequency of use, and collaboration.

3 Research Setting

We used the Google Forms tool to create an online survey. The poll was anonymous, although the questionnaire included some general questions about demographic data that described the respondent's background and workplace. The most crucial inquiries, however, focused on the respondents' expertise in Agile Requirement Elicitation Techniques. The survey included questions regarding 16 agile requirement elicitation techniques based on three parameters i.e., Ease of Use, Frequency, and Collaboration. The survey was carried out by experienced persons from the agile industry. There were 35 responses. The survey's completion rate was estimated to be 100%. Table 2 presents some important facts about the respondents' backgrounds. About 14% had limited experience (Less than 1 year), 68% declared experience between 1 and 5 years, and roughly 14% claimed more than 10 years in agile development.

We built this article on discussions and problems from prior research, as well as data acquired by practitioners. It resulted in the following research questions:

RQ1: What are the requirement elicitation techniques used in agile?

RQ2: Prioritize requirement elicitation techniques used in agile based on ease of use, frequency of use, and collaboration.

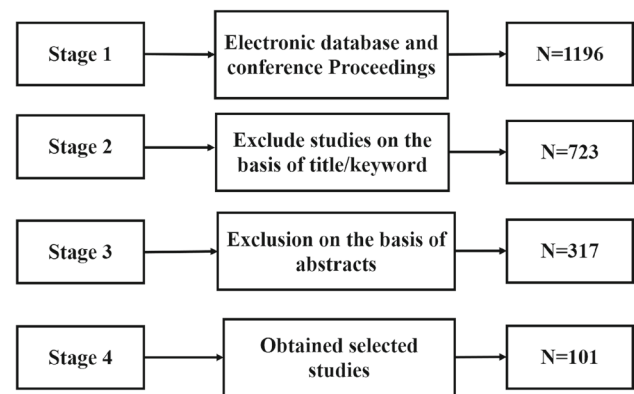


Fig. 1 Study selection process

To fill this gap, we will address these research questions in this paper, by prioritizing agile requirements elicitation techniques based on fuzzy logic.

4 Research Methodology

4.1 Inclusion and Exclusion Criteria

A thorough literature review was done in order to prioritize the agile requirement elicitation techniques. The following five well-known electronic databases were used in this review: IEEE Xplore, Elsevier Science Direct, ACM Digital Library, Google Scholar, and Springer Link. They covered sufficient literature for the study.

In Fig. 1, each stage of the review process is identified. Table 3 shows the terms and keywords used for search in the first stage. These databases looked sufficient for the review. The keywords mentioned below are searched on these different databases in which some of them were used and searched as a phrase by using inverted commas (") and with the use of the "AND" operator we make sure that all the possible combinations in the search process of desired literature should be done.

The total results of the database searches were 1196 and then we started the exclusion process on the basis of irrelevant titles and keywords and we shortlisted 723 articles. Then, the shortlisted articles were evaluated on the basis of Abstracts and relevancy. We shortlisted an article list of 317 and on the basis of deep understanding and further study, we shortlisted and finalized 101 articles.

4.2 Agile Requirements Elicitation Techniques

Schon2017 Agile conducted a literature review on software requirements elicitation techniques. Based on the work, 57 different techniques/artifacts were identified. The artifacts which were only referred to once were excluded from the

Table 1 Related work

Authors	Description of research	Technique	Evaluation measures	Limitations
Raslan et al. [24]	The study describes a fuzzy logic-based framework integrating COCOMO cost drivers and story points to enhance effort estimation through EEEM and IEEM	COCOMO II	MRE,MAE and PRED	Accuracy depend on defined fuzzy set and rules with,possible improvement using different member functions
Tayh et al. [25]	The authors showed how combining user stories with a systematic approach to factor quantification might improve the accuracy of work estimation using fuzzy logic	fuzzy logic technique	Accuracy, comparison, validation	The study does not employ a systematic approach to quantifying factors and lacks comparison with actual data from software projects
Jawale et al. [26]	The study proposed Adaptive FHCV that uses adaptive mechanism with FHCV to prioritize requirements	Fuzzy logic technique	Comparative Analysis	Adaptive fuzzy hierarchical cumulative voting takes more time in prioritization
Bakly et al. [27]	This research uses fuzzy logic to enhance the hybrid technique for prioritization to control uncertainty	Hybrid Model with fuzzy logic	Comparative between Classical and Fuzzy Technique	Achieved 50% of success in ranking the requirements
Winton et al. [28]	This paper presents the comparative analysis of fuzzy AHP and fuzzy TOPSIS for requirement selection	Fuzzy AHP, Fuzzy TOPSIS	Comparative analysis	Fuzzy AHP gives less accurate results. More disagreements on ranking
Nazim et al. [29]	This paper proposes interactive prioritization approach to rank requirements using constraint solver and pairwise comparisons	Constraint Satisfaction	Experiments, Validation	Fuzzy AHP give no accurate priority and create conflicts. Although fuzzy TOPSIS gives consistent results
Hassan et al. [30]	This study introduces social networks and fuzzy-based method is applied or requirement elicitation	Fuzzy logic, Social networks	Case Study	Limited to institute examination system, not evaluated on real time systems
Ala et al. [31]	In this research, authors have introduced fuzzy methods to address the effects of uncertain parameters along with other optimization algorithms	Fuzzy logic	Case Study, Comparison	Route optimization can be more effective by using machine learning algorithms and blockchain for end-to-end visibility
Abusaeed et al. [32]	This study has identified the cost factors for project failures and prioritized these factors using FAHP technique to handle uncertainty	Fuzzy AHP	Comparison	It is yet to find the most effective cost estimation technique by using MCDM

Table 2 Demographic information about respondents

Answers	No. of responses	% of responses
<i>Experience in agile development</i>		
Less than 1 year	5	14.2
1–5 years	24	68.5
6–10 years	1	2.8
More than 10 years	5	14.2

Table 3 Search terms

Search category	Keywords
Agile	Agile elicitation, XP requirements
Fuzzy	Fuzzy logic, scrum using fuzzy, prioritization using fuzzy
Requirement elicitation	Agile requirement elicitation, requirements gathering

list. The artifacts that were mentioned at least twice made the list and are shown in Table 4.

4.3 Prioritization Parameters: Agile Requirements Elicitation Techniques

There can be many prioritization parameters, while we try to prioritize any list. We followed a systematic approach to identify and shortlist the prioritization parameters. Following is the sequence of activities performed to identify and shortlist the prioritization parameters. a. Based on SLR articles, we identify an initial list of prioritization parameters. b. On the basis of frequency, we shortlisted three parameters. Table 5 shows the parameters finalized.

4.4 Application of Fuzzy Logic

4.4.1 What is Fuzzy?

Fuzzy refers to a lack of clarity, and this fuzziness is due to the employment of a complex mathematical pattern to model the most similar humanitarian inference. A fuzzy system basically converts human knowledge into mathematical formulas. Multi-valued logic can be seen as an extension of fuzzy logic. Its purposes and goals, on the other hand, are entirely different. Fuzzy logic works with approximate rather than precise modes of reasoning as a result, in fuzzy logic, the chains of reasoning are typically short, and consistency is less significant than it is in classical logical systems. In general, a degree determines everything in fuzzy logic, including truth.

4.4.2 Why Fuzzy is Used?

The idea of a linguistic variable and the underlying logic of fuzzy if-then rules is the most obvious, well-understood, and commonly utilized contribution to fuzzy logic. Fuzzy logic

can best interpret the human language that are imprecise and uncertain. As expert language and reasoning are vague then fuzzy logic is well suited. It allows us to mimic the responses of experts gathered through survey to model the prioritization of techniques. Researchers have utilized fuzzy logic in combination with other techniques for prioritization. Some of the papers combined techniques by using different types of hybrid techniques, such as Adaptive Fuzzy Hierarchical Cumulative Voting (AFHCV), hybrid Gamified method with AHP, FuzzyHCV -hybrid of HCV, and Fuzzy Expert System, which focus on the role of stakeholders in the process of requirements [26,27].

4.4.3 How Fuzzy is Applied?

The first step is to gather survey from the practitioners about requirement elicitation techniques. Then, we convert the linguistic terms into the corresponding weight of importance using fuzzy logic to handle uncertainty. This way we prioritize the techniques based on the shortlisted parameters.

4.5 Collecting the Importance Weights of the Requirements Elicitation Techniques and its Parameters

We use triangular fuzzy numbers (TFNs) because we identified three potential parameters; ease of use, frequency of use, and collaboration for prioritization, and experts gave responses on the basis of these parameters. As we are combining three fuzzy numbers, TFN makes it easier to combine these parameters and model their contributions for overall evaluation. Recently, authors have utilized the similar approach of fuzzy methods for evaluation of agile patterns [33] and for evaluation of coordination and communication processes in offshore software development [34].

Using TFNs, the study demonstrated a simple way to understand linguistic terms and parameters to show subjec-

Table 4 Artifacts used for agile requirements elicitation

Sr.	Artifacts	Description
T1	User story	A user story is a feature description that is written from the user's perspective It consists of written text, discussion of the text, and standards for acceptability
T2	Prototype	A software application paradigm called prototype helps in the assessment and communication of design concepts
T3	Use case	A use case is a sequence of activities required to accomplish a specified goal
T4	Scenario	A scenario is a written depiction of an issue that describes the user-system interaction in a specific situation
T5	Story Card	The textual text and information of a user story are physically represented on a story card
T6	Persona	A persona is a fictional character who represents a big portion of the target audience
T7	Vision	A vision is referred to as the overarching goal that directs product development and unites development, business professionals, and other stakeholders
T8	UML Diagram	For modeling a system's design, there is a standard called the Unified Modeling Language (UML)
T9	Storyboard	The user's workflow is shown in a series of photos using a storyboard
T10	Kanban Board	The workflow of the development team is shown on the Kanban board to show how a requirement is progressing
T11	UI Pattern	A user interface pattern is an abstract solution to reoccurring design difficulties that serves as inspiration for designers

Table 4 continued

Sr.	Artifacts	Description
T12	Essential Use Cases	A use case that outlines user tasks can be summarized and generalized into an essential use case
T13	Pictures	Any visual representation of something is called a picture
T14	Videos	A video is a collection of images that have undergone electronic processing, are captured on film, and are displayed on a screen
T15	Mind Map	It is a visual representation of knowledge that helps you visualize and organize it
T16	UI Specifications	A written specification that outlines a system's user interface. Mock-ups, icons, and other visual elements are used to enhance the text

Table 5 Shortlisted prioritization parameters

S. nos.	Prioritization parameters
1	Ease of use
2	Frequently used
3	Collaboration

tive endorsement for or objection toward certain requirements elicitation techniques. Because not all techniques are equally valuable, we were interested in determining the importance of weight. A range of zero to one is assigned to memberships to objects within a fuzzy set's universe of discourse. Let U be a universal set with u as its elements, then the membership function of a fuzzy set X is

$$\mu_x(u) \rightarrow U[0, 1] \quad (1)$$

which assigns a membership grade X to each u in the range $[0, 1]$.

Several articles state that applying fuzzy set theory to human cognition can decrease subjective fuzziness [35–37]. A language scale was offered as a practical way of unfolding in such situations. We used a seven-point linguistic scale to assign importance weight to Requirement Elicitation Techniques, as shown in Table 6.

Table 6 Linguistic terms for weighting techniques

Linguistic terms	Corresponding weight of importance
Extremely agree	(0.9, 1.0, 1.0)
Moderately agree	(0.7, 0.9, 1.0)
Slightly agree	(0.5, 0.7, 0.9)
Neutral	(0.3, 0.5, 0.7)
Slightly disagree	(0.1, 0.3, 0.5)
Moderately disagree	(0.0, 0.1, 0.3)
Strongly disagree	(0.0, 0.0, 1.0)

Similarly, the respondents are given seven linguistic characteristics to rate the use of Requirements Elicitation Techniques across three parameters (ease of use, frequency of use, collaboration), as shown in Table 7, based on research [38]. The experts were requested to give their views about each technique with respect to the three parameters. Outstanding shows that the technique is performing well given the parameters and Very Poor indicates that the technique is not the best performing across these parameters.

4.6 Establishing the Extent to which Agile Requirement Elicitation Techniques are Prioritized

The following steps explain the procedures for prioritizing agile requirement elicitation techniques as described in [34]:

Step 1: Build matrix A to translate the responses of the survey respondents using the scale as shown in Table 7.

$$A = \begin{bmatrix} \alpha_1^1 & \alpha_1^2 & \alpha_1^3 & \dots & \alpha_1^n \\ \alpha_2^1 & \alpha_2^2 & \alpha_2^3 & \dots & \alpha_2^n \\ \alpha_3^1 & \alpha_3^2 & \alpha_3^3 & \dots & \alpha_3^n \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \alpha_p^1 & \alpha_p^2 & \alpha_p^3 & \dots & \alpha_p^n \end{bmatrix} \quad (2)$$

Table 7 Linguistic terms for rating techniques

Linguistic terms	Corresponding weight of importance
Outstanding	(0.9, 1.0, 1.0)
Fully implemented	(0.7, 0.9, 1.0)
Marginally implemented	(0.5, 0.7, 0.9)
Fair	(0.3, 0.5, 0.7)
Weak	(0.1, 0.3, 0.5)
Poor	(0.0, 0.1, 0.3)
Very poor	(0.0, 0.0, 1.0)

where p represents the total number of techniques and n represents the total number of respondents, $\alpha_p^n = (l\alpha_p^n, m\alpha_p^n, u\alpha_p^n)$ shows the fuzzy weight of the technique given by the n th respondent for p th technique. One example, of the result, is given in Table 7.

Step 2: Due to the fact that each respondent's subjective assessment of their experience, role, perception, and comprehension of the subject matter differs. Therefore, we incorporated the mean score approach to aggregate the fuzzy importance of each technique by n respondent. By this way, we interpret the average of expert opinion about each technique.

$$w_p = \frac{1}{n} \left[\sum_{i=1}^n \alpha_p^i \right] \quad (3)$$

where $w_p = (lw_p, mw_p, uw_p)$ shows the aggregate fuzzy importance weight of the p th technique.

Step 3: The aggregated TFN_{w_p} is used to obtain the best non-fuzzy performance (BNF) value, BNP_{w_p} . BNP_{w_p} can be produced using Eq. (4).

$$BNP_{w_p} = \frac{[(uw_p - lw_p) + (mw_p - lw_p)]}{3} + lw_p \quad (4)$$

We defuzzify the TFN to get the crisp numeric value to do ranking. Here, BNP_{w_p} represents the BNP value for the TFN_{w_p} , while w_p is the important weight of the p th technique in classical (crisp) number format..

Step 4: After the defuzzification of TFN in step 4, crisp numbers are obtained and normalized using Eq. (5). We normalized the crisp numbers to make sure that they all stay within the range [0, 1].

$$R_p = \frac{w_p}{\sum_{p=1}^n w_p} \quad (5)$$

where R_p shows the normalized significance weight of the p th technique such that $\sum_{p=1}^n R_p = 1$

Table 8 Corresponding TFNs (weighting) of techniques

Experts	Corresponding TFNs (weighting) of techniques T1–T8								
	D	T1	T2	T3	T4	T5	T6	T7	T8
E1	E	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.9,1.0, 1.0)
	F	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.9,1.0, 1.0)
	C	(0.9, 1.0, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.9,1.0, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.9,1.0, 1.0)
	M	(0.9, 1.0, 1.0)	(0.76, 0.93, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.76, 0.93, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.9,1.0, 1.0)
E2	E	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)
	F	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)
	C	(0.9, 1.0, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)
	M	(0.76, 0.93, 1.0)	(0.9,1.0, 1.0)	(0.83, 0.96, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)
E3	E	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C	(0.9, 1.0, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)
	M	(0.76, 0.93, 1.0)	(0.9,1.0, 1.0)	(0.76, 0.93, 1.0)	(0.76, 0.93, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.76, 0.93, 1.0)
E4	E	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)
	F	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)
	C	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.9,1.0, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	M	(0.36, 0.56, 1.0)	(0.43, 0.63, 0.83)	(0.83, 0.96, 1.0)	(0.56, 0.76, 0.9)	(0.43, 0.63, 0.83)	(0.36, 0.56, 0.76)	(0.33, 0.5, 0.7)	(0.23, 0.43, 0.63)
E5	E	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.9,1.0, 1.0)
	C	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.9,1.0, 1.0)
	M	(0.76, 0.93, 1.0)	(0.83, 0.96, 1.0)	(0.76, 0.93, 1.0)	(0.76, 0.93, 1.0)	(0.7, 0.9, 1.0)	(0.36, 0.56, 0.76)	(0.43, 0.63, 0.8)	(0.9,1.0, 1.0)
E6	E	(0.9,1.0, 1.0)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
	F	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)
	C	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)
	M	(0.83, 0.96, 1.0)	(0.43, 0.63, 0.83)	(0.23, 0.43, 0.63)	(0.3, 0.5, 0.7)	(0.36, 0.56, 0.76)	(0.16, 0.36, 0.56)	(0.23, 0.43, 0.63)	(0.23, 0.43, 0.63)
E7	E	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)
	F	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)
	C	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)
	M	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)
E8	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)
	F	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
	C	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
	M	(0.9, 1.0, 1.0)	(0.76, 0.93, 1.0)	(0.7, 0.9, 1.0)	(0.83, 0.96, 1.0)	(0.76, 0.93, 1.0)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)

Table 8 continued

Experts	Corresponding TFNs (weighting) of techniques T1–T8							
D	T1	T2	T3	T4	T5	T6	T7	T8
E9	E (0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	F (0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	C (0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
	M (0.63, 0.83, 0.96)	(0.7, 0.9, 1.0)	(0.56, 0.76, 0.93)	(0.56, 0.76, 0.93)	(0.7, 0.86, 0.96)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.56, 0.76, 0.93)
E10	E (0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
	F (0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
	C (0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
	M (0.3, 0.5, 0.7)	(0.23, 0.43, 0.63)	(0.36, 0.56, 0.76)	(0.36, 0.56, 0.76)	(0.43, 0.63, 0.83)	(0.16, 0.36, 0.56)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
E11	E (0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)
	F (0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)
	C (0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	M (0.5, 0.7, 0.86)	(0.76, 0.93, 1.0)	(0.83, 0.96, 1.0)	(0.7, 0.9, 1.0)	(0.56, 0.76, 0.93)	(0.7, 0.9, 1.0)	(0.83, 0.96, 1.0)	(0.83, 0.96, 1.0)
E12	E (0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)
	F (0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.0, 0.0, 1.0)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)
	C (0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)
	M (0.83, 0.96, 1.0)	(0.43, 0.63, 0.83)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.06, 0.23, 0.43)	(0.33, 0.46, 0.93)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)
E13	E (0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)
	F (0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)
	C (0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)
	M (0.16, 0.36, 0.56)	(0.03, 0.16, 0.36)	(0.13, 0.3, 0.5)	(0.06, 0.23, 0.43)	(0.3, 0.5, 0.7)	(0.03, 0.16, 0.36)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)
E14	E (0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)
	F (0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.0, 0.0, 1.0)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)
	C (0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)
	M (0.1, 0.3, 0.5)	(0.33, 0.5, 0.7)	(0.03, 0.1, 0.83)	(0.2, 0.33, 0.56)	(0.3, 0.5, 0.7)	(0.23, 0.43, 0.63)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)
E15	E (0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F (0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C (0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	M (0.76, 0.93, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
E16	E (0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	F (0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	C (0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	M (0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.83, 0.96, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)

Table 8 continued

Experts	Corresponding TFNs (weighting) of techniques T1–T8							
D	T1	T2	T3	T4	T5	T6	T7	T8
E17	E (0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)
	F (0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)
	C (0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)
	M (0.63, 0.83, 0.96)	(0.36, 0.56, 0.76)	(0.43, 0.63, 0.83)	(0.36, 0.56, 0.76)	(0.43, 0.63, 0.83)	(0.36, 0.56, 0.76)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)
E18	E (0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	F (0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)
	C (0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)
	M (0.56, 0.76, 0.9)	(0.63, 0.8, 0.93)	(0.63, 0.8, 0.93)	(0.43, 0.63, 0.83)	(0.56, 0.76, 0.93)	(0.43, 0.63, 0.8)	(0.5, 0.7, 0.9)	(0.56, 0.73, 0.86)
E19	E (0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)
	F (0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)
	C (0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)
	M (0.5, 0.7, 0.86)	(0.56, 0.76, 0.93)	(0.56, 0.76, 0.93)	(0.63, 0.8, 0.9)	(0.1, 0.3, 0.5)	(0.06, 0.23, 0.43)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)
E20	E (0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
	F (0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
	C (0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
	M (0.7, 0.86, 0.96)	(0.9, 1.0, 1.0)	(0.63, 0.83, 0.96)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
E21	E (0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F (0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)
	C (0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)
	M (0.43, 0.63, 0.83)	(0.5, 0.7, 0.83)	(0.7, 0.9, 1.0)	(0.63, 0.83, 0.96)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.76, 0.93, 1.0)	(0.7, 0.9, 0.96)
E22	E (0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F (0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)
	C (0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	M (0.56, 0.76, 0.9)	(0.63, 0.7, 0.76)	(0.56, 0.73, 0.86)	(0.56, 0.76, 0.93)	(0.63, 0.83, 0.96)	(0.5, 0.66, 0.8)	(0.76, 0.93, 1.0)	(0.63, 0.83, 0.96)
E23	E (0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.0, 0.1, 0.3)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)
	F (0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)
	C (0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)
	M (0.63, 0.83, 0.96)	(0.76, 0.9, 0.96)	(0.56, 0.73, 0.83)	(0.63, 0.8, 0.9)	(0.56, 0.73, 0.86)	(0.0, 0.1, 0.3)	(0.43, 0.63, 0.8)	(0.83, 0.96, 1.0)
E24	E (0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)
	F (0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)
	C (0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)
	M (0.63, 0.8, 0.9)	(0.23, 0.43, 0.63)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)

Table 8 continued

Experts	Corresponding TFNs (weighting) of techniques T1–T8							
	T1	T2	T3	T4	T5	T6	T7	T8
E25	E	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	M	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.63, 0.83, 0.96)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.86)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)
E26	F	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)
	C	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)
	M	(0.83, 0.96, 1.0)	(0.56, 0.76, 0.93)	(0.63, 0.76, 0.83)	(0.76, 0.93, 1.0)	(0.36, 0.56, 0.76)	(0.7, 0.9, 1.0)	(0.43, 0.6, 0.73)
	E	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)
	F	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)
E27	C	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)
	M	(0.63, 0.8, 0.93)	(0.5, 0.7, 0.86)	(0.56, 0.76, 0.9)	(0.7, 0.9, 1.0)	(0.43, 0.63, 0.83)	(0.76, 0.9, 0.96)	(0.3, 0.5, 0.7)
	E	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)
	F	(0.3, 0.5, 0.7)	(0.0, 0.0, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)
	C	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
E28	M	(0.56, 0.73, 0.86)	(0.4, 0.5, 0.9)	(0.76, 0.93, 1.0)	(0.3, 0.5, 0.7)	(0.43, 0.56, 0.8)	(0.83, 0.96, 1.0)	(0.5, 0.7, 0.9)
	E	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)
	F	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)
	C	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)
	M	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.13, 0.3, 0.5)	(0.0, 0.03, 0.76)	(0.0, 0.06, 0.53)	(0.0, 0.1, 0.3)	(0.0, 0.06, 0.53)
E29	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	F	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	C	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	M	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
E30	F	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	C	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	M	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	F	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
E31	C	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	M	(0.43, 0.63, 0.8)	(0.56, 0.76, 0.9)	(0.63, 0.8, 0.9)	(0.56, 0.76, 0.9)	(0.03, 0.16, 0.36)	(0.03, 0.16, 0.36)	(0.23, 0.43, 0.63)
	E	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)
	F	(0.1, 0.3, 0.5)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)
	C	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.7, 0.9, 1.0)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)
E32	M	(0.03, 0.16, 0.36)	(0.0, 0.0, 1.0)	(0.4, 0.56, 0.73)	(0.03, 0.16, 0.36)	(0.26, 0.43, 0.6)	(0.16, 0.36, 0.56)	(0.43, 0.63, 0.8)

Table 8 continued

Experts	Corresponding TFNs (weighting) of techniques T1–T8								
	D	T1	T2	T3	T4	T5	T6	T7	T8
E33	E	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)
	F	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)
	C	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)
	M	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.03, 0.76)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.03, 0.16, 0.36)	(0.06, 0.23, 0.43)	(0.03, 0.16, 0.36)
E34	E	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
	C	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
	M	(0.76, 0.93, 1.0)	(0.7, 0.9, 1.0)	(0.83, 0.96, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.56, 0.76, 0.93)
E35	E	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	F	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	C	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	M	(0.36, 0.56, 0.76)	(0.7, 0.86, 0.96)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
E1	E	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	M	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.56, 0.76, 0.93)	(0.56, 0.76, 0.93)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
E2	E	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	M	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
E3	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	M	(0.9, 1.0, 1.0)	(0.83, 0.96, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
E4	E	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)
	C	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
	M	(0.4, 0.56, 0.73)	(0.36, 0.56, 0.76)	(0.3, 0.5, 0.7)	(0.43, 0.63, 0.8)	(0.43, 0.63, 0.8)	(0.43, 0.63, 0.8)	(0.56, 0.76, 0.9)	(0.56, 0.76, 0.93)

Table 8 continued

Experts	Corresponding TFNs (weighting) of techniques T1–T8								
	D	T1	T2	T3	T4	T5	T6	T7	T8
E5	E	(0.9,1.0, 1.0)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)
	F	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.7, 0.9, 1.0)
	C	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.9,1.0, 1.0)	(0.9,1.0, 1.0)
	M	(0.56, 0.73, 0.86)	(0.36, 0.56, 0.76)	(0.7, 0.9, 1.0)	(0.9,1.0, 1.0)	(0.56, 0.76, 0.93)	(0.56, 0.76, 0.93)	(0.9,1.0, 1.0)	(0.76, 0.93, 1.0)
E6	E	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)
	F	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)
	C	0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)
	M	(0.16, 0.36, 0.56)	(0.1, 0.3, 0.5)	(0.23, 0.43, 0.63)	(0.06, 0.23, 0.43)	(0.03, 0.16, 0.36)	(0.1, 0.3, 0.5)	(0.06, 0.23, 0.43)	(0.06, 0.23, 0.43)
E7	E	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)
	F	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)
	C	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)
	M	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)
E8	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)
	F	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)
	C	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)
	M	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)
E9	E	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
	M	(0.63, 0.83, 0.96)	(0.7, 0.9, 1.0)	(0.63, 0.83, 0.96)	(0.63, 0.83, 0.96)	(0.76, 0.93, 1.0)	(0.83, 0.96, 1.0)	(0.63, 0.83, 0.96)	(0.7, 0.9, 1.0)
E10	E	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)
	F	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
	C	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)
	M	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.36, 0.56, 0.76)	(0.3, 0.5, 0.7)	(0.36, 0.56, 0.76)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
E11	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
	C	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)
	M	(0.83, 0.96, 1.0)	(0.7, 0.86, 0.96)	(0.63, 0.83, 0.96)	(0.83, 0.96, 1.0)	(0.7, 0.86, 0.96)	(0.76, 0.9, 0.96)	(0.83, 0.96, 1.0)	(0.56, 0.76, 0.93)

Table 8 continued

Experts	Corresponding TFNs (weighting) of techniques T1–T8								
	D	T1	T2	T3	T4	T5	T6	T7	T8
E12	E	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.9, 1.0, 1.0)
	F	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.0, 0.0, 1.0)	(0.5, 0.7, 0.9)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)
	C	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.5, 0.7, 0.9)
	M	(0.76, 0.93, 1.0)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.0, 0.06, 0.53)	(0.5, 0.7, 0.9)	(0.0, 0.0, 1.0)	(0.0, 0.0, 1.0)	(0.46, 0.56, 0.96)
E13	E	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)
	F	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)
	C	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)
	M	(0.23, 0.43, 0.63)	(0.43, 0.63, 0.83)	(0.03, 0.16, 0.36)	(0.13, 0.3, 0.5)	(0.03, 0.16, 0.36)	(0.03, 0.16, 0.36)	(0.03, 0.16, 0.36)	(0.13, 0.3, 0.5)
E14	E	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)	(0.0, 0.1, 0.3)
	F	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)	(0.0, 0.1, 0.3)
	C	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)	(0.0, 0.1, 0.3)
	M	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)	(0.0, 0.1, 0.3)
E15	E	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	M	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
E16	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	F	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	C	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	M	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.63, 0.8, 0.93)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)

Table 8 continued

Experts	Corresponding TFNs (weighting) of techniques T1–T8							
D	T1	T2	T3	T4	T5	T6	T7	T8
E17	E	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	F	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
	C	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
	M	(0.5, 0.7, 0.9)	(0.43, 0.63, 0.83)	(0.5, 0.7, 0.86)	(0.36, 0.56, 0.76)	(0.7, 0.9, 1.0)	(0.56, 0.76, 0.9)	(0.63, 0.83, 0.96)
E18	E	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
	F	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)
	C	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)
	M	(0.43, 0.63, 0.83)	(0.43, 0.63, 0.83)	(0.5, 0.7, 0.86)	(0.63, 0.83, 0.96)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.5, 0.66, 0.8)
E19	E	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)
	C	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)
	M	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.43, 0.63, 0.83)	(0.4, 0.56, 0.73)	(0.26, 0.43, 0.63)	(0.63, 0.83, 0.96)
E20	E	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	F	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	C	(0.7, 0.9, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	M	(0.76, 0.93, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.76, 0.93, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
E21	E	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)
	F	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.9, 1.0, 1.0)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)
	C	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	M	(0.26, 0.43, 0.6)	(0.36, 0.56, 0.76)	(0.5, 0.7, 0.86)	(0.7, 0.9, 0.96)	(0.46, 0.63, 0.76)	(0.26, 0.43, 0.6)	(0.5, 0.7, 0.86)
E22	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.5, 0.7, 0.9)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	C	(0.1, 0.3, 0.5)	(0.9, 1.0, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
	M	(0.5, 0.66, 0.8)	(0.6, 0.7, 0.76)	(0.63, 0.8, 0.93)	(0.5, 0.66, 0.8)	(0.43, 0.63, 0.83)	(0.56, 0.76, 0.93)	(0.63, 0.83, 0.96)
E23	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
	F	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	M	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.76, 0.93, 1.0)	(0.56, 0.76, 0.9)	(0.56, 0.76, 0.93)	(0.56, 0.76, 0.93)
E24	E	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	C	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)
	M	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)

Table 8 continued

Experts	Corresponding TFNs (weighting) of techniques T1–T8								
	D	T1	T2	T3	T4	T5	T6	T7	T8
E25	E	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	M	(0.7, 0.9, 1.0)	(0.63, 0.83, 0.96)	(0.7, 0.9, 1.0)	(0.63, 0.83, 0.96)	(0.63, 0.83, 0.96)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
E26	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
	F	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
	C	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.3, 0.5, 0.7)
	M	(0.76, 0.9, 0.96)	(0.76, 0.9, 0.96)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.43, 0.63, 0.8)	(0.36, 0.56, 0.76)	(0.56, 0.76, 0.9)
E27	E	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)
	F	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)
	C	(0.9, 1.0, 1.0)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)
	M	(0.7, 0.86, 0.96)	(0.3, 0.5, 0.7)	(0.7, 0.9, 1.0)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.43, 0.63, 0.83)	(0.3, 0.5, 0.7)	(0.63, 0.83, 0.96)
E28	E	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.7, 0.9, 1.0)
	F	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.7, 0.9, 1.0)
	C	(0.3, 0.5, 0.7)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)
	M	(0.36, 0.56, 0.76)	(0.3, 0.5, 0.7)	(0.36, 0.56, 0.76)	(0.43, 0.63, 0.83)	(0.36, 0.56, 0.76)	(0.16, 0.35, 0.56)	(0.23, 0.43, 0.63)	(0.7, 0.9, 1.0)
E29	E	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)
	F	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)
	C	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)
	M	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)
E30	E	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	F	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	C	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
	M	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)	(0.9, 1.0, 1.0)
E31	E	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)
	F	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)
	C	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)
	M	(0.16, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.16, 0.36, 0.56)	(0.0, 0.1, 0.3)	(0.03, 0.16, 0.36)
E32	E	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.5, 0.7, 0.9)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)
	F	(0.1, 0.3, 0.5)	(0.5, 0.7, 0.9)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.7, 0.9, 1.0)	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)
	C	(0.0, 0.0, 1.0)	(0.3, 0.5, 0.7)	(0.0, 0.1, 0.3)	(0.3, 0.5, 0.7)	(0.5, 0.7, 0.9)	(0.3, 0.5, 0.7)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)
	M	(0.03, 0.13, 0.6)	(0.36, 0.56, 0.76)	(0.03, 0.13, 0.6)	(0.03, 0.13, 0.6)	(0.36, 0.56, 0.76)	(0.5, 0.7, 0.86)	(0.06, 0.2, 0.66)	(0.03, 0.16, 0.36)

Table 8 continued

Experts	Corresponding TFNs (weighting) of techniques T1–T8								
	D	T1	T2	T3	T4	T5	T6	T7	T8
E33	E	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)
	F	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)
	C	(0.1, 0.3, 0.5)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)
	M	(0.03, 0.16, 0.36)	(0.1, 0.3, 0.5)	(0.0, 0.1, 0.3)	(0.03, 0.16, 0.36)	(0.03, 0.16, 0.36)	(0.0, 0.1, 0.3)	(0.0, 0.0, 1.0)	(0.0, 0.1, 0.3)
E34	E	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
	C	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
	M	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)	(0.7, 0.9, 1.0)	(0.5, 0.7, 0.9)
E35	E	(0.9, 1.0, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	F	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	C	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)
	M	(0.76, 0.93, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)	(0.7, 0.9, 1.0)

Table 9 Importance weight and possible ranking of techniques

Techniques#	wj = (l, m, u)			BNP_WJ	Rj	Over all ranking
T1	0.55	0.72	0.87	0.71	0.067171	2
T2	0.54	0.69	0.87	0.71	0.067171	3
T3	0.57	0.73	0.9	0.73	0.069063	1
T4	0.51	0.68	0.85	0.68	0.064333	4
T5	0.56	0.64	0.81	0.67	0.063387	6
T6	0.41	0.57	0.76	0.58	0.054872	16
T7	0.48	0.65	0.82	0.65	0.061495	8
T8	0.53	0.68	0.83	0.68	0.064333	5
T9	0.51	0.53	0.84	0.62	0.058657	15
T10	0.51	0.66	0.79	0.65	0.061495	7
T11	0.49	0.66	0.82	0.65	0.061495	9
T12	0.48	0.64	0.84	0.65	0.061495	10
T13	0.47	0.65	0.84	0.65	0.061495	11
T14	0.47	0.63	0.82	0.64	0.060549	14
T15	0.44	0.6	0.85	0.65	0.061495	12
T16	0.52	0.65	0.78	0.65	0.061495	13
BNP_WJ					10.57	

5 Discussion

Tables 8 and 9, respectively, provide the important weight and potential ranking of the 16 requirement elicitation techniques based on the three prioritization factors: ease of use, frequently used, and collaboration. In our findings, T3, T1, T2, T4, and T8 are techniques widely implemented in relation to the overall importance weight shown in Table 9. Based on our fuzzy ranking, the respondents need to follow the following techniques for requirement elicitation in an effective way. Based on how easy to use the technique was, how frequently it was used by practitioners, and how a technique encourages collaboration. T3: Use Case was ranked 1 as 40% participants claimed that it encourages collaboration. It aids in defining functional requirements clearly within the boundary or scope of the project. T1: User story was ranked 2 as user stories facilitate in describing requirements from the user's point of view in a written format T2: Prototyping was ranked 3 as 37% of practitioners agreed that prototyping helps in the evaluation and sharing of design ideas of a project through collaboration. T4: Scenarios were ranked 4 as practitioners describe scenarios as a description of user-system interaction in a specific situation. 35% of practitioners agreed that scenarios are easy to use. T8: UML diagrams were ranked 5 as they provide a standard for visualizing a system's design. 31% of practitioners agree that UML diagrams are easy to use and 31% agree that they are currently frequently used for requirement elicitation. T5: Story card was ranked 6, as practitioners agree that they show the physical description of user stories in a textual and detailed format. 29% of practitioners agree user cards are easy to use and 23% agree to story cards

are a technique that encourages collaboration among the team in the requirement process. T10: Kanban board was ranked 7 as nowadays most companies use agile methodologies for their projects and Kanban boards can be used to illustrate the progress of a requirement via the development team's workflow. 26% of practitioners agree that Kanban boards are easy to use and are frequently used for requirement elicitation. 23% agree that Kanban boards facilitate team collaboration. T7: Vision was ranked 8 as it guides product development and aligns the development of business personnel. 23 % of practitioners agreed that visions are easy to use and encourage collaboration among the team. T11: UI Patterns was ranked 9 as they provide the abstract solution to reoccurring design difficulties. 40% of the practitioners agreed that UI Patterns encourage 40% team collaboration. T12: Essential Use Cases were ranked 10 as they facilitate in describing user tasks. 31% agree that Essential Use Cases encourage team collaboration. T13: Pictures were ranked 11 as they provide a visual representation of some form of requirements. 29% of the practitioners agree that pictures are frequently used for requirement elicitation and 34% agree that they encourage collaboration. T15: Mind Map was ranked 12 as they are used for visual representation of knowledge regarding requirement elicitation. 25% of the practitioners agreed that Mind Map was easy to use and 23% agreed that it was frequently used and encouraged collaboration. T16: UI Specifications were ranked 13 as they outline a system's user interface for requirement elicitation. 34 % of the practitioners agreed that UI Specifications encourage team collaboration T14: Videos were ranked 14 as they provide a visual representation of requirements. 23% agree that Videos are frequently used for

requirement elicitation and 33% agree that they encourage team collaboration. T9: Storyboard was ranked 15 as it shows the user's workflow in a series of photos using a storyboard. 32% agree that Storyboard facilitates collaboration that is required for requirement elicitation. T6: Persona ranked 16 as it represents users' requirements through functional character. 11% of the practitioners agreed it was easy to use, 20% agreed that it was frequently used, and 14% agreed that it facilitates team collaboration.

6 Threat to Validity

There are four main types of threats to validity, which are internal validity, external validity, construct validity, and conclusion validity. We conducted a threat to the validity of our results. While identifying the requirement elicitation techniques, the research focused only on the commonly used 11 techniques including Kanban, Use Cases, Pictures, etc. We did not focus on techniques that are not popularly used as the focus of the research was to prioritize the most used techniques. This can pose a threat to validity as there may be new requirement elicitation techniques that do not fall into the criteria.

For the fuzzy set, three factors were selected: ease of use, frequently used, and collaboration. This can pose a threat as well as it limits the results. Other researchers can find more factors that affect the prioritization of the requirement elicitation techniques. For the questionnaire, a limited number of participants were selected for the sample set. This can pose a threat to validity as the selection of research participants was limited by the willingness of experts to participate and the availability of experts in the domain of software development. As with any empirical software engineering project, a very high number of variables can affect the project.

7 Conclusion and Future Work

In conclusion, this research proposed a fuzzy-based approach to prioritize agile requirement elicitation techniques. To answer the research questions, a survey was conducted based on the data collected from the survey fuzzy weights were applied. T3: Use Case was ranked as the first priority based on our criteria as 40% of practitioners agree that Use Cases are mostly used for collaboration which answered the RQ2. This will facilitate partitioners to use Use Cases for their requirement process. Similarly, regarding RQ1 the result shows that the 16 requirement elicitation techniques are potentially used in agile development. This list will serve as a catalog for practitioners to select requirement elicitation techniques for their projects.

Future work can focus on studying more criteria other than just ease of use, frequent use, and collaboration for prioritizing requirement elicitation techniques as currently our research focuses on them. Additionally, a tool-based solution can also be developed that could help practitioners select which requirement elicitation techniques are suited for their projects.

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